King Fahd University of Petroleum & Minerals
Information and Computer Science Department

ICS 507: Design and Analysis of Parallel Algorithms
Project
(Due: Saturday May 2, 2026 at Midnight)

In this project, we will study the problem of parallelization of matrix multiplication, with respect to Algorithm PARMATRIXMULT and parallelizing Strassen's Divide-and-Conquer algorithm. The sequential straightforward way of carrying out the matrix multiplication is given below.

```
/* Sequential Version */
void multiply(int A[][N], int B[][N], int C[][N])
{
  for (int i = 0; i < N; i++) {
     for (int j = 0; j < N; j++) {
          C[i][j] = 0;
          for (int k = 0; k < N; k++) {
               C[i][j] += A[i][k]*B[k][j];
          }
     }
  }
}
```

The project will be carried out in phases.

Phase 1 (10 points): Background (Due Saturday April 11, 2026 at midnight)

In this phase, you will study and report on Strassen's Algorithm. First, you will describe Strassen's divide and conquer algorithm and write the algorithm. Second, you will analyze its time complexity, in terms of the number of scalar multiplications. Make sure to justify the superior performance of Strassen's algorithm over the straightforward sequential algorithm.
The outcome of this phase is a report on the above consisting of 2-3 pages.

Phase II [70 points]: Implementation, Experimentation and Analysis (Due Saturday May 2, 2026 at midnight)

Question 1 [20 points] Program Implementation

Use OpenMP to develop

1. (2 points) The sequential multiplication version.
2. (4 points) A parallelization of the sequential multiplication version using Algorithm PARMATRIXMULT in the textbook.
3. (4 points) The sequential Strassen's divide and conquer algorithm.
4. (10 points) The Parallelized Strassen's divide and conquer algorithm.

Your implementation needs to take the following considerations into account:

- The input should be read from a text file containing the following information
  - The matrix dimension n in the first line. The input matrices are going to be $n \times n$ matrices, where n is a power of 2.
  - $2n^2$ integer values, entered in a row-major form starting from the second line onward, representing the two input matrices, respectively. Each input value is a random integer between -9 and 9, inclusive.
- The values should all be of type "long integer", so that you do not run into overflow

problems due to multiplying matrices of large sizes.

- Your code should be able to support matrices of sizes up to $2^{14} \times 2^{14}$ (or as far as you can get).
- There shall be eight output text files generated, two for each multiplication method:
  - The first file is the output resulting matrix. The filename shall consist of the input filename, followed by the dimension of the output matrix n followed by the word "output" followed by the method of multiplication. Please use the following for the name of the multiplication method: *Sequential, ParMtrixMult, Strassen, and ParStrassen*.
    For example, if the input filename is input1 and the dimension of the matrices is 128×128, then the output files containing the results of the multiplication for each method are:
    - input1-128-output-Sequential.txt
    - input1-128-output-ParMtrixMult.txt
    - input1-128-output-Strassen.txt
    - input1-128-output-ParStrassen.txt
  - The second file contains the time (hh:mm:ss) taken by the matrix multiplication algorithm followed by the number of cores used. Note that you need to exclude the time of reading the input or writing the output. The filename shall consist of the input filename, followed by the dimension of the output matrix n followed by the word "info" followed by the method of multiplication.

Question 2 [40 points] Experimental Evaluation

Run the four implementations on different matrices with varying parameters of dimensions, number of processors used, and base cases for the stopping criteria of Strassen's algorithm (both sequential and parallel). Tabulate your results in tabular form and give your assessment of the four algorithms based on your experimentation.

Question 3 [10 points]

Analyze the performance of each parallel algorithm theoretically by providing the following information:

1. The shared memory model used for each parallel implementation.
2. The expected running time of each parallel implementation as a function of the size n and the number of processors p.
3. Explain any discrepancies that you may encounter due to difference between the theoretical analysis and the actual performance of each implementation.

Phase III (20 points): Presentation

In this phase, an online presentation of each project will be given, which will be attended by ALL STUDENTS. The presentation should not exceed 10 minutes and should have the following information:

a. Status of the project: This includes what was implemented and what was not implemented. What was additionally implemented but not required, if any. Any limitations of your implementations.
b. Implementation details of the two parallel algorithms.
c. Description of the experimentation and summary of the results. This includes the specifications of the machine used (RAM, # Cores, CPU Cache, Speed, etc.).
d. Conclusions.

The presentations will be conducted in the last two lectures of the semester, in shaa Allaah.

IMPORTANT NOTES REGARDING THE DELIVERABLES

1. This project will be done in groups of 3 students.
2. **Project Phase I** is to be submitted on Gradescope. Make sure you map the submission to the outline of the assignment.
3. **Project Phase II Code** is to be submitted on Gradescope as a zip file containing the following directories:
    a. src: Your Project Implementation [Answer to **Question 1**].
    b. doc: A short README file on how to compile and run your code.
4. **Project Phase II Report** is to be submitted on Gradescope, mapping it to the following outline:
    a. Cover page.
    b. Status of the project. This includes
        i. What has been implemented from the required algorithms.
        ii. Mention any additional algorithms implemented, if any.
        iii. Details of the implementation of the two parallel algorithms.
        iv. What are the limitations, if any, of your implementation[s] and or difficulties encountered.
        v. Who did what in the project.
    c. Experimentation Results
        i. Specification of the machine used in running the algorithms.
        ii. How many runs were performed, what size of matrices were used, …etc. [Answer to **Question 2**]
        iii. Analysis of the performance of all algorithms. [Answer to **Question 3**]
    d. Summary and Conclusion
    e. Any references used in the development of the parallel algorithms or preparing the background.
5. **Project Phase III Presentation** is to be submitted on Blackboard.